Paper 6.50 - Causal Code Claim Contract

CQE-paper-06.50
CQECMPLX Formal Paper Series
Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-06.50.md

Paper 6.50 - Causal Code Claim Contract

Purpose

Paper 6.50 defines what counts as a valid causal-code claim. It ensures that proof dependencies, review loops, open obligations, and rejected paths remain distinguishable.

Admitted Paper 6 Claims

The following claims are admitted from Paper 6:

```
valid causal edges require source, target, edge_type, receipt, status
allowed edge types are uses, proves, refines, obligates, transports, repairs, constrains,
verifies
allowed statuses are open, closed, deferred, rejected
closed proof-support edges must be acyclic
open obligations remain open
missing receipt, unknown type, and hidden proof cycle are rejected
```

Claim Requirements

Any later paper using causal code must state:

```
which dependency is being represented
which edge type is used
which receipt supports the edge
which status the edge has
whether the edge belongs to closed proof support or review/obligation routing
```

Linked Receipt

The minimum receipt link for Paper 6 is:

```
paper: CQE-paper-06
theorem: Typed causal edge contract
receipt: production/formal-papers/CQE-paper-06/causal_code_receipt.json
status: pass
```

The receipt is sufficient for the typed dependency theorem. It is not sufficient by itself to claim the full 32-paper graph is populated or all obligations are closed.

Boundary Failures

The following are boundary failures:

```
using a dependency without a receipt
inventing an edge type without updating the verifier
counting open obligations as closed
hiding proof cycles under prose
treating administrative metadata as proof without graph validation
```

Boundary failures are rejected or routed to obligations.

Hidden-Guess Contract

When hidden-guess diagnostics are enabled, the answer must be recorded before the receipt is revealed. The guess record should include:

```
prompt
predicted edge validity or graph status
revealed receipt
match/mismatch
next obligation if mismatch
```

The hidden-cycle prompt is required because it tests whether the reviewer can spot circular support before the verifier reveals the result.

Conclusion

Paper 6.50 lets later papers import causal code honestly. It keeps the graph useful as proof infrastructure without allowing open or circular support to be mistaken for closure.